

Experiment: 6

5G NR Frame Structure Analysis with Different Subcarrier Spacing

Objective – 1: Analyze Different Subcarrier Spacing Values

```
clc;
```

```
clear;
```

```
close all;
```

```
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
```

```
mu = 0:4; % Numerology Index
```

```
bar(scs)
```

```
grid on
```

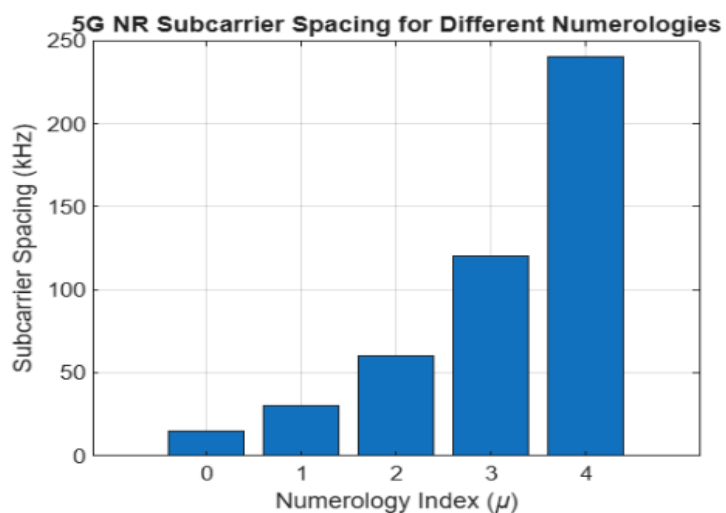
```
set(gca,'XTick',1:5)
```

```
set(gca,'XTickLabel',mu)
```

```
xlabel('Numerology Index ( $\mu$ )')
```

```
ylabel('Subcarrier Spacing (kHz)')
```

```
title('5G NR Subcarrier Spacing for Different Numerologies')
```



Objective – 2 Compare Slot Duration

```
clc;
```

```
clear;
```

```
close all;
```

```
% Subcarrier Spacing (kHz)
```

```
scs = [15 30 60 120 240];
```

```
% Slot Duration (ms)
```

```
slot = [1 0.5 0.25 0.125 0.0625];
```

```
% Plot
```

```
figure;
```

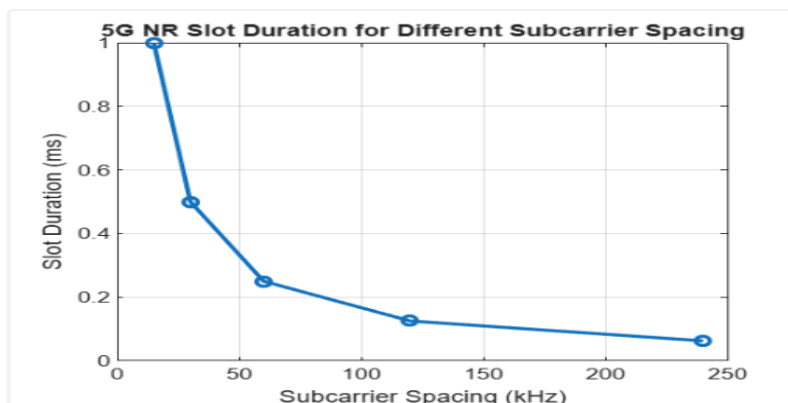
```
plot(scs, slot, 'o-', 'LineWidth', 2);
```

```
grid on;
```

```
xlabel('Subcarrier Spacing (kHz)');
```

```
ylabel('Slot Duration (ms)');
```

```
title('5G NR Slot Duration for Different Subcarrier Spacing');
```



Objective – 3 Study the Number of OFDM Symbols

`clc;`

`clear;`

`close all;`

`scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)`

`symbols = [14 14 14 14 14]; % OFDM Symbols per Slot`

`figure`

`bar(scs, symbols)`

`grid on`

`xlabel('Subcarrier Spacing (kHz)')`

`ylabel('Number of OFDM Symbols')`

`title('OFDM Symbols per Slot for Different Subcarrier Spacing')`

